

Retail Behavioural Analytics

An end-to-end video analytics pipeline over two CCTV clips from a mall shoe shop: store interest and conversion at the entrance, per-shelf interest inside, and staff–customer interaction sessions.

Everything runs locally. No footage leaves the machine.

1. Results

Task 1 — Store interest and conversion	Total Interested	14
	Interested Entered	10
	Interested Passed By	4
Task 2 — Per-shelf interest events	A / B / C / D	6 / 4 / 3 / 2 (15 total)
Task 3 — Staff interaction	Average sessions per staff instance	0.57 (4 sessions ÷ 7 instances)

Deliverables: `outputs/entrance_annotated.mp4`, `outputs/interior_annotated.mp4`, `outputs/task1_store_interest.csv`, `outputs/task2_shelf_interest.csv`, `outputs/task3_staff_interactions.csv`, and the full decision record in `outputs/events/*.json`.

How well does it work? Scored against 39 blind hand labels for Task 1:

	precision	recall	F1
all 39 labels	0.80	0.50	0.62
the 27 labels the reviewer marked <i>confident</i>	1.00	0.60	0.75

On the labels the reviewer was confident about there are no false positives at all: every person the pipeline calls interested, is. Both hand-verified entries are detected (2 of 2). The failures are mostly the other way — people who show interest briefly without sustaining it past the threshold.

Four of these labels were **corrected during review**: tracks 65, 76, 81 and 536 were originally recorded as entering the shop, and frame-by-frame checking showed all four stay outside the shopfront and leave at the right edge of frame. The correction and its reason are recorded in `labels/task1_labels.csv`.

Read §4.4 before putting weight on those two rows. The labels were produced by the candidate, and §4.4 shows that a label set produced this way could not distinguish a working orientation estimator from one that was guessing. They are reported because they are the only interest labels that exist, not because they are a good instrument.

Staff detection against 10 hand-verified staff instances: **precision 1.00, recall 0.70**. Every disagreement is named in §9.

2. Setup and running

Docker (recommended — reproduces every number with no GPU)

```
docker build -t rva .
docker run --rm -v "$PWD/outputs:/app/outputs" rva analyze # ~30 s, no GPU
docker run --rm rva test # 64 tests

# render also needs the source clips, which are not baked into the image
docker run --rm -v "$PWD/raw_videos:/app/raw_videos" \
    -v "$PWD/outputs:/app/outputs" rva render # ~4 min
```

`analyze` regenerates all three CSVs and both event files from the committed `cache/tracks_*.parquet`. It needs no GPU and no network. That is deliberate — see §3.

Local

```
python -m venv .venv && . .venv/Scripts/activate # Windows
pip install --index-url https://download.pytorch.org/whl/cu130 \
    torch==2.13.0+cu130 torchvision==0.28.0+cu130 # or the cpu index
pip install -r requirements.txt
```

Model weights: `yolo11m-pose.pt` downloads automatically on first use, or

```
python -c "from ultralytics import YOLO; YOLO('yolo11m-pose.pt')"
```

CLIP (`openai/clip-vit-base-patch32`) downloads from the Hugging Face hub on first use, and is only needed to *reproduce the staff-classifier experiment* in §7 — the shipped pipeline does not use it.

```
# Pass 1 - GPU, ~7 min per clip. Only needed to regenerate cache/.
python scripts/run_perception.py entrance interior

# Pass 2 - CPU, seconds.
python scripts/analyze.py

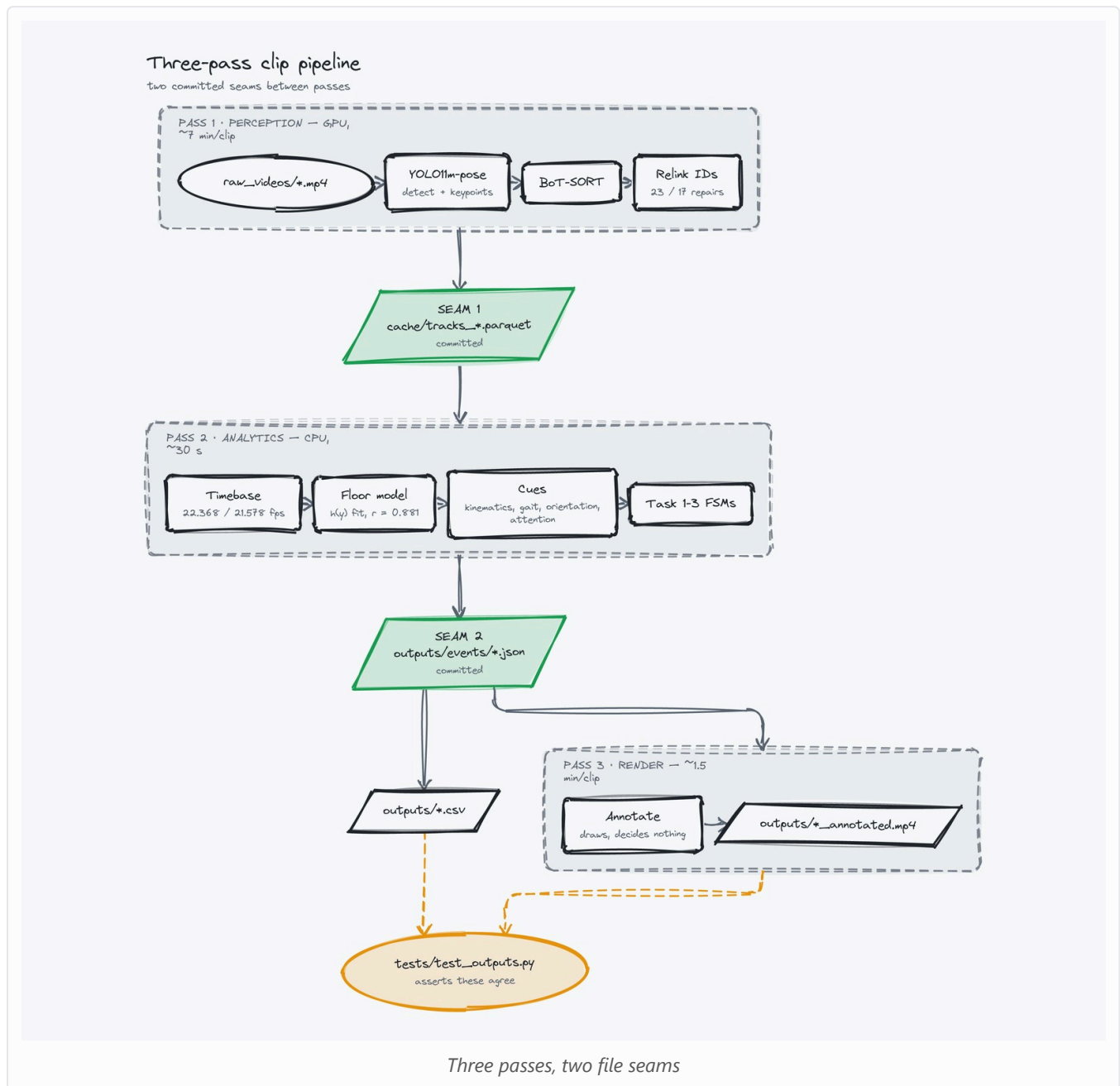
# Pass 3 - CPU, ~3 min.
python scripts/run_render.py entrance interior # ~3 min total

python -m pytest tests/ -q # 64 tests
```

Probes (each produces the evidence for a number quoted below)

```
python scripts/probe_timebase.py      --scene entrance    # the frame-rate lie
python scripts/calibrate.py           --scene entrance    # the floor model
python scripts/preview_zones.py       --scene entrance    # zones over a density map
python scripts/probe_orientation.py   --scene entrance --selfcheck # label-free, preferred
python scripts/probe_orientation.py   --scene entrance --score      # hand labels; see 4.4
python scripts/build_staff_crops.py   --scene entrance
python scripts/rank_staff_candidates.py --scene entrance
python scripts/train_staff.py          --scene entrance --variant both
python scripts/fit_task1.py
python scripts/review_task1.py        --score
```

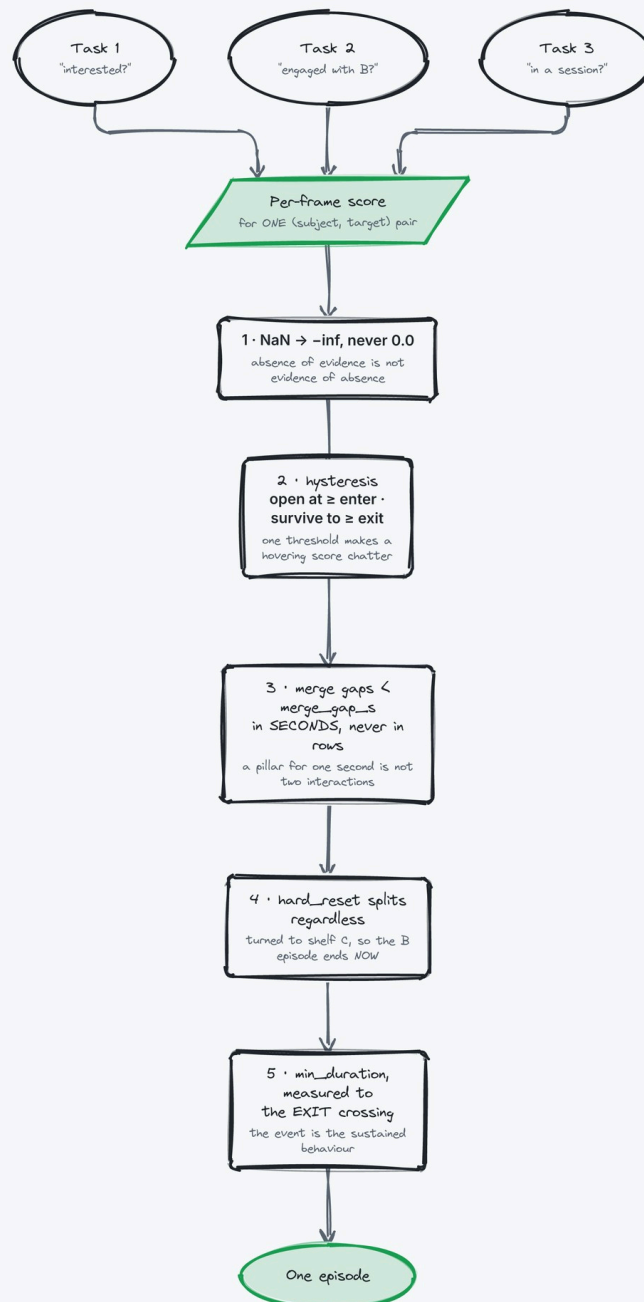
3. Architecture



Three passes, two file seams. Only Pass 1 needs a GPU; the parquet is committed, so every number regenerates on CPU in about thirty seconds.

One episode

three tasks, one tested rule for what "one event" means



One counting rule, shared by all three tasks

All three questions are the same question: given a noisy per-frame score for one (subject, target) pair, how many distinct sustained episodes occurred? Solving it once, in one tested module, is why the three tasks cannot disagree about what "one event" means.

Three passes, two file seams.

```

raw_videos/*.mp4
|
▼ Pass 1 · perception    GPU, ~7 min/clip    scripts/run_perception.py
cache/tracks/*.parquet    ← COMMITTED to the repo
|
▼ Pass 2 · analytics    CPU, seconds        scripts/analyze.py
outputs/events/*.json → outputs/*.csv
|
▼ Pass 3 · render        CPU, ~1.5 min/clip  scripts/run_render.py
outputs/*_annotated.mp4

```

Why the seams are where they are. Detection and tracking are the only stage that needs a GPU and cannot be reproduced bit-for-bit elsewhere. Everything downstream is deterministic CPU arithmetic. Committing the parquet means a reviewer with no GPU regenerates every number in this README in about thirty seconds, and it means a threshold change costs seconds rather than eleven minutes.

Pass 3 decides nothing. Every count, episode and verdict drawn on the video is read from `outputs/events/*.json` — the same file the CSVs are formatted from. The renderer does compute presentation detail (an elapsed timer on an active session, a running total that ticks up as episodes start), but it applies no threshold and derives no episode, so there is no second implementation of any rule for the video and the CSV to disagree about. `tests/test_outputs.py` asserts the equality rather than trusting it.

```

src/rva/
  config.py          typed, fail-fast scene config (pydantic, extra="forbid")
  timebase.py        the true frame rate
  io/                tracks.py (schema)  events.py  csvout.py
  perception/        detect_track.py  relink.py  embed.py
  geometry/          calibrate.py  floor.py  zones.py
  features/          kinematics.py  gait.py  orientation.py  attention.py
  staff/             dataset.py  colour.py  model.py  infer.py
  analytics/         fsm.py  task1_entrance.py  task2_shelf.py  task3_staff.py
  render/            draw.py  annotate.py  encode.py

```

Models and libraries. YOLO11m-pose (Ultralytics, AGPL-3.0) for detection, tracking (BoT-SORT) and 17-point pose in a single forward pass; shapely for zone geometry; scikit-learn for the classifier experiment; pydantic for config; ffmpeg/libx264 for encoding. One model, because pose costs nothing extra over plain detection and everything behavioural here needs keypoints.

4. The findings that shaped the pipeline

4.1 The container lies about the frame rate



First and last frame of the entrance clip. The container claims 30 fps; the clock burnt into the corner spans exactly 600 s across 13,421 frames, which is 22.368.

`cv2.CAP_PROP_FPS` reports **30.0**. The burnt-in CCTV clock advances from `15:27:17` to `15:37:17` — exactly **600.0 s** — across 13,421 frames.

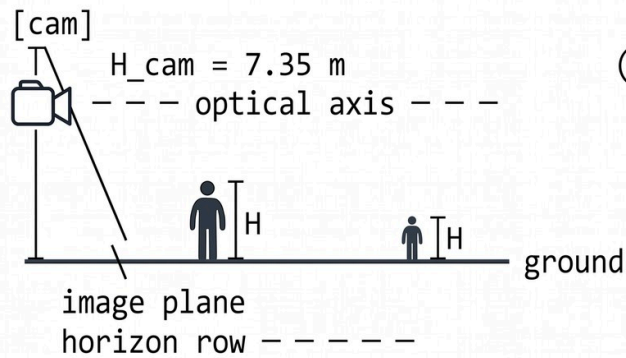
	entrance	interior
container says	30.0 fps → 447 s	30.0 fps → 432 s
clock says	600.0 s	600.0 s
true rate	22.368 fps	21.578 fps
header error	+34.1 %	+39.0 %

Every rule in this pipeline is written in seconds. Believing the header would make every duration threshold 34 % too short and nothing would visibly break — the pipeline would just report confident, wrong counts.

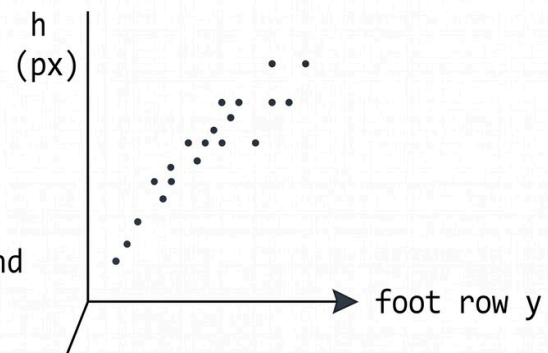
`docs/figures/timebase_entrance.png` shows the two clock readings side by side so this can be checked by eye rather than taken on trust.

4.2 The floor model, from people as rulers

LEFT - side elevation



RIGHT - the fit



$y_{\text{horizon}} = -226$ • line crosses $h = 0$ here

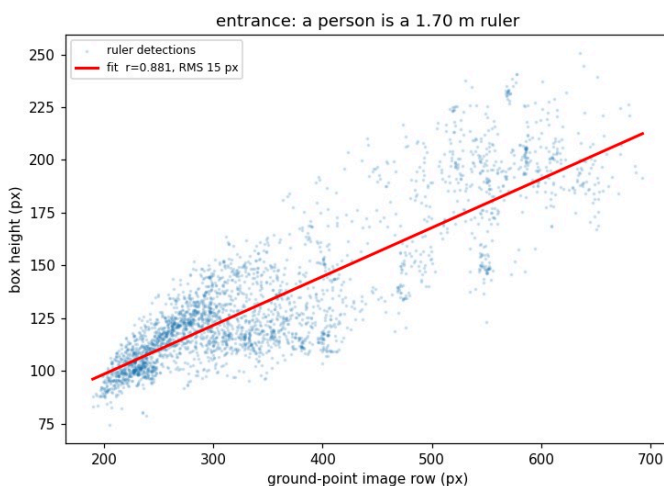
$$h(y) = (H / H_{\text{cam}}) \cdot (y - y_{\text{horizon}})$$

$r = 0.881$ • residual 14.76 px • 2,891 samples • max 40 per track

slope and horizon are MEASURED
absolute scale rests on $H = 1.70 \text{ m}$ and HFOV 100°

Pixel height against foot row

A standing person is a ruler that walks around. Pixel height is linear in foot row and reaches zero at the horizon.



Calibration fit and a reprojected one-metre grid

Left, the fit over 2,891 samples: $r = 0.881$, residual 14.76 px. Right, a one-metre floor grid reprojected onto a frame — the check that does not use the fit's own residual.

There is no calibration target and no survey. But adult standing height is roughly constant, so a person is a ruler that walks around. For a level pinhole camera, a standing person of height H whose feet are at image

row *y* has pixel height $h(y) = (H / H_{cam}) \cdot (y - y_{horizon})$ — linear, through zero at the horizon.

	entrance	interior
rulers used	2,891	2,055
height-vs-row fit	$r = \mathbf{0.881}$, RMS residual 14.8 px	$r = \mathbf{0.964}$, RMS 10.6 px
horizon row	−226	−184
implied camera height	7.35 m	4.93 m

The slope and horizon are **measured**. Only the absolute scale rests on the two assumptions (stature 1.70 m, HFOV 100°); the relative perspective — how much bigger a metre is near the camera than far away, which is what every comparison downstream depends on — does not.

One vote per person per position, not one per frame. A person standing in view for 40 s contributes ~900 near-identical rows. Capping each track's contribution at 40 evenly-spaced samples moved *r* from 0.832 → 0.881, the residual from 19.2 → 14.8 px, and the implied camera height from an implausible 8.23 m to 7.35 m. 900 rows of one stationary person is one measurement repeated, not 900 measurements.

External checks, which do not depend on the fit:

check	measured	expected
mall walkway	12.8 m × 3.2 m	a real corridor
shopfront width	10.3 m	a real shopfront
shelf footprints	2.4 / 2.8 / 3.5 / 2.8 m wide	real retail fixtures
p90 of per-track median walking speed	1.20 (entrance) / 1.06 (interior) m/s	published 1.2–1.4 m/s

Two independently fitted cameras landing on the low edge of the published walking-speed band is the strongest evidence available that the scale is right.

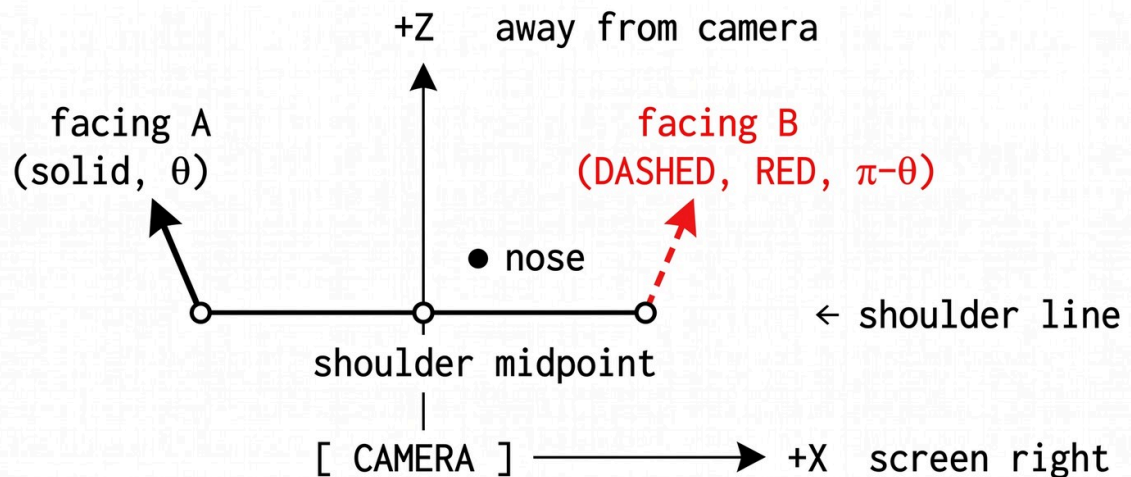
4.3 Gait cadence validates the timebase, and caught a bug

Ankle separation, normalised by box height, peaks once per step. Counting prominent peaks gives steps per second:

	measured	expected
entrance, while walking (>0.6 m/s)	1.72 steps/s (IQR 1.32–2.03)	human band 1.6–2.4
entrance, while stationary	0.39 steps/s	≈0
interior, while walking	1.08 steps/s	below band — see §9

The first implementation counted zero crossings of the detrended signal and reported ~4 steps/s, roughly twice the human band. Keypoint jitter adds several extra crossings per step. Counting *prominent peaks* — did the legs actually separate and come back together, or did two dots wobble — fixed it. The known biomechanical constant is what made the bug visible at all.

4.4 Orientation, the bug in it, and why the first validation missed it



what the camera sees
 shoulder offset $\propto -u_z$
 IDENTICAL for both arrows $\Rightarrow$ two valid solutions

the tie-breaker

 nose right of shoulder centre
 $\Rightarrow$ faces screen-right
 95 % agreement · 83 % coverage

before	117° median	55 % reversed
after	28° median	5 % reversed

The two-fold ambiguity and the nose that breaks it

The shoulder offset constrains toward-or-away from the camera and says nothing about left versus right, so every side-on frame has two equally good solutions. The nose breaks the tie: 117° and 55 % reversed, down to 28° and 5 %.

The brief defines interest as "looking toward the storefront, turning their head or body toward it". Turning toward something you are walking *past* means facing and travelling are different directions — so an estimator that assumes people face where they walk cannot detect the behaviour being asked for.

The geometry, and the degeneracy hiding in it. Write facing on the floor as a unit vector $\mathbf{u} = (u_x, u_z)$, X across the image and Z away from the camera. The shoulder line is perpendicular to $\mathbf{u}$, so projecting it gives a horizontal shoulder offset in the image proportional to $-u_z$. That constrains whether the person faces toward or away from the camera — and says *nothing* about the sign of u_x , whether they face screen-left or screen-right. Every frame admits two solutions, θ and $\pi - \theta$, that fit the shoulders equally well.

The first implementation swept 72 candidate yaws and took the argmin of a cost built from the shoulder offset and the pattern of visible facial keypoints. Both terms constrain `uz`. **Neither constrains the sign of `ux`**. So the two mirror solutions had near-identical cost and the argmin picked whichever grid cell noise favoured — a coin flip, exactly when the person was side-on.

How that was found, and why it was nearly missed. The original validation was 40 hand-labelled crops, and it reported pose at 50° median error with 8% near-reversed — comfortably beating direction of travel. That result was wrong, and the reason is worth more than the number: the labels and the estimator were produced by the same person, so when a small blurry crop was misread it was misread *the same way* the model misread it, and the errors cancelled.

The check that caught it needs no labels at all. People walking briskly are, with very few exceptions, facing the way they are going, so for fast movers the direction of travel is a free reference of thousands of frames that nobody authored. Direction of travel was itself verified against raw pixel motion first (median 0° error), so it does not rest on the model it is testing.

entrance, brisk walkers	median error	near-reversed
original estimator , shoulders in profile	150°	55%
original estimator , all	117°	40%
repaired estimator , shoulders in profile	29°	2%
repaired estimator , all	28°	5%

The confidence was the second failure: it sat at ~0.5 whether the estimator knew the answer or was guessing.

The repair — one signal per unknown.

- `uz` comes from the *signed* shoulder offset. The pose model labels shoulders anatomically, so a camera-facing person has their left shoulder on the image right. Verified on this footage: of 10,521 detections with nose and both eyes confident, **100%** have `lsx > rsx`; of 683 showing ears but no nose, **0%**.
- the sign of `ux` comes from where the **nose** sits relative to the shoulder midpoint. Someone in profile facing screen-right has their nose right of their shoulder centre. On exactly the profile frames where the shoulders are blind: **95% agreement, 83% coverage**. Ears were tried and are much weaker (60%, 85%).
- with no nose, the sign falls back to direction of travel, marked lower confidence; with neither, `yaw_conf` goes to zero and the cues abstain.
- the confidence now scales the penalty for a weak sign source by `|ux|` — by how much the sign actually *matters* on that frame. Getting it wrong costs nothing square-on and 180° in profile.

The confidence does not model the whole error, and the self-check says so. Broken out by shoulder spread, the worst bin is *square-on* — 41 deg median and 11% reversed, against 22-29 deg in the profile bins — and it simultaneously carries the **highest** mean confidence, 0.90. That inversion is expected from the construction above: confidence penalises a weak left/right sign in proportion to `|ux|`, and `|ux|` goes to zero square-on, so the sign penalty vanishes exactly there. Nothing in the confidence expresses how well *the angle itself* is conditioned, and square-on is where it is conditioned worst: spread varies as roughly `|cos yaw|`, whose derivative vanishes at the square-on extremum, so a fixed pixel error maps to an unbounded yaw error.

Two things could produce that 41 deg and this run does not separate them: the ill-conditioning just described, and the fact that the *reference* is weakest in the same bin — a brisk walker who is square-on is moving mostly in depth, where direction of travel is recovered from the floor model and is itself noisy. The test that would separate them is to restrict the comparison to frames whose travel direction is well determined in the image plane. I have not run it, so the honest statement is that 41 deg is an upper bound on the estimator's error in that bin, not a measurement of it.

This is a known limitation rather than a live defect: `facing` and `head_turn` both carry weight **0.0** in the fitted Task 1 model (section 5.2), so orientation contributes nothing to the headline entrance numbers. It feeds Task 2 only, through the angular alignment term, and there it is multiplied by a confidence that is too generous square-on. Fixing it means folding a conditioning term such as `|sin yaw|` into `yaw_conf`, which would shift the Task 2 counts; I chose not to change a verified number on the strength of an effect I have not yet isolated.

Reproduce with `python scripts/probe_orientation.py --scene entrance --selfcheck`.

The hand-label probe is kept, and demoted. Run against the *repaired* estimator it reports 52° — statistically the same as the 50° it gave the broken one, while the label-free check moved 117° → 28°. That is the clearest possible demonstration that the instrument was blind, and it is why `--selfcheck` is now the primary validation and `--score` carries a warning.

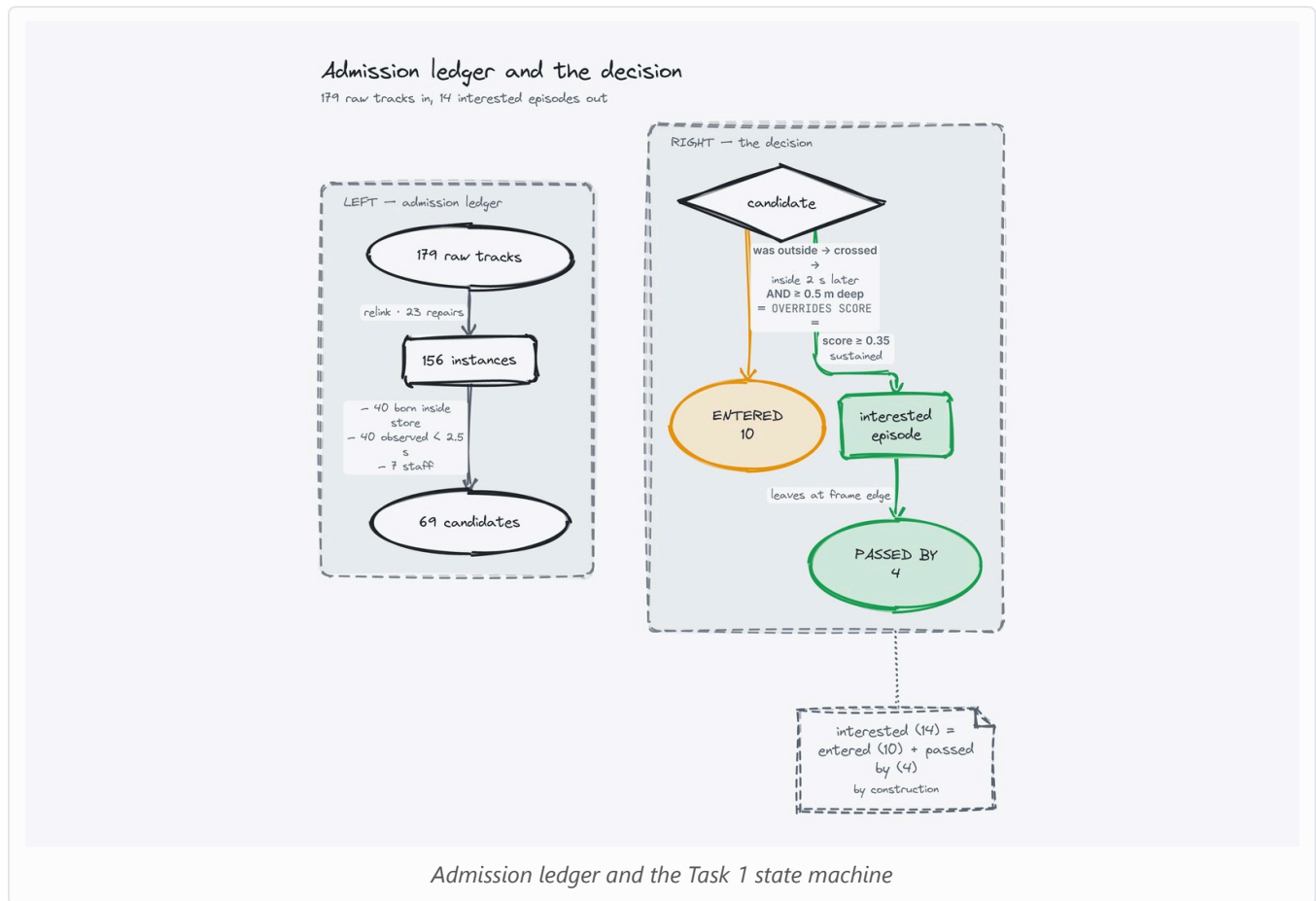
4.5 The walkway is only two metres deep, which changes what distance means

Measured from the ground-point distribution: the visible walkway band spans floor depth 6.9–9.0 m while the shopfront sits at 6.7–7.9 m. **Every person on the walkway is between 0.04 m and 1.25 m from the frontage** (5th–95th percentile).

So distance has to be resolved at the scale of about a metre. A 2.5 m proximity radius — an intuitively reasonable choice — would score every walkway person identically and carry no information at all. The proximity cues use `full 0.3 m → zero 1.5 m`, and the reliability decay uses `1.5 m → 6.0 m`.

5. Task 1 — Store interest and conversion

5.1 The admission ledger



Admission first, then interest. Entry is decided by geometry and overrides the cue score, so $\text{entered} \subseteq \text{interested}$ holds by construction rather than by luck.



Left, the density of all 51,000 ground points in the clip — what the shopfront line was traced against, rather than a single still.

Before counting interest, decide who could possibly be an interested passer-by.

exclusion	n	why
born inside the store	40	already in the shop; walking <i>out</i> is not passer-by interest
observed < 2.5 s	40	cannot distinguish a person from a tracker artefact
staff	7	see §7
admitted candidates	69	

The ledger is order-dependent by design, and that shows in these counts. Under an earlier, larger store polygon the split was 95 / 15 / 7 → 39 candidates: tracks that now fail the duration test were previously absorbed by the born-inside test first. The total is always accounted for.

Order matters. Store-polygon membership at the person's *first* observation is tested before anything else. A customer leaving the shop passes a naive "was outside" test on their way out and would otherwise be counted as an interested passer-by. `tests/test_outputs.py` asserts the reasons sum to the exclusion count.

5.2 Cues, and how they are weighted

Each cue is continuous in [0, 1]; NaN means *no evidence*, and the blend renormalises over the cues that have evidence, so a missing cue **abstains** rather than voting against.

Weights are proportional to (AUC - 0.5), normalised, against the 39 hand labels -- one measured statistic per cue rather than six free parameters, because six free parameters on 39 samples fit the noise and generalise to nothing.

cue	what it measures	AUC	coverage	weight
<code>approaching</code>	closing on the shopfront	0.734	56 %	0.408
<code>dwelling</code>	sustained slowness near the window	0.625	10 %	0.218
<code>hesitating</code>	cadence to 0 while near the window	0.619	15 %	0.208
<code>slowing</code>	below 0.7x the person's own median pace	0.595	23 %	0.165
<code>head_turn</code>	facing the store while <i>not</i> walking toward it	0.458	23 %	0
<code>facing</code>	oriented at the nearest point of the frontage	0.347	54 %	0

With 8 positives among 39 labels the standard error on each AUC is about 0.11, so only `approaching` is clearly distinguishable from chance and the ordering of the rest should not be over-read.

A finding this table refuted, which is worth more than the table. An earlier version of this README reported `slowing` at 0.22 and `dwelling` at 0.21 -- significantly *anti*-correlated with entering -- and drew a behavioural conclusion from it: that people who linger at a window are the ones who walk on. After the shop zones were re-traced (section 6) those same cues score 0.60 and 0.63, mildly positive. The "finding" was an artefact of a store boundary in the wrong place. It is recorded here because a plausible behavioural story is exactly the kind of error that survives review.

`facing` and `head_turn` **falling to zero is uncomfortable and is reported rather than hidden.** They are the behaviours the brief names first, and the orientation estimator behind them is separately validated at 28 degrees median error (section 4.4). On this clip, against these labels, body orientation does not discriminate

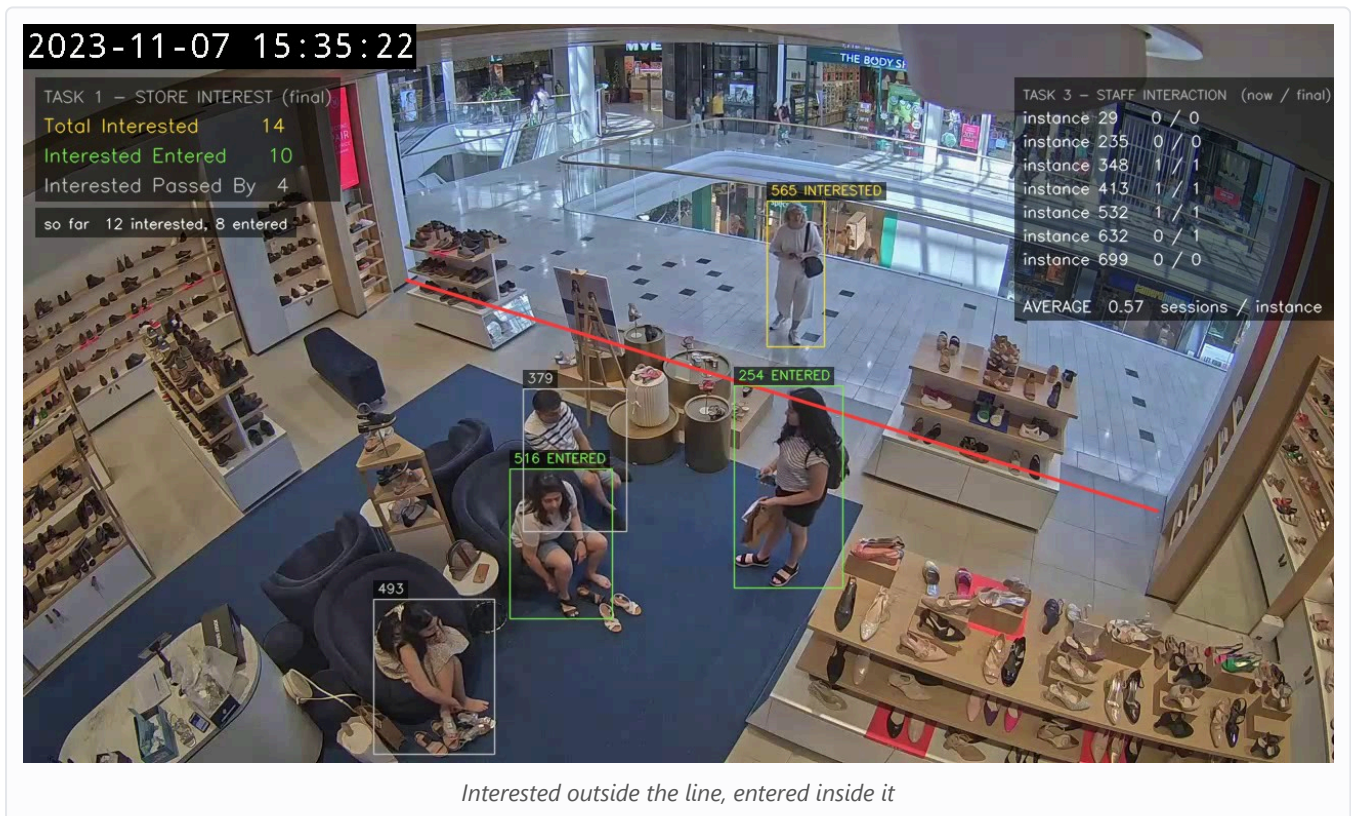
interest at the storefront. It remains load-bearing for Task 2, where it decides which shelf a customer is facing.

Two alternatives were tried on the same labels and are worse: a stated prior over the brief's named behaviours (precision 0.36) and equal weights across all six (0.32).

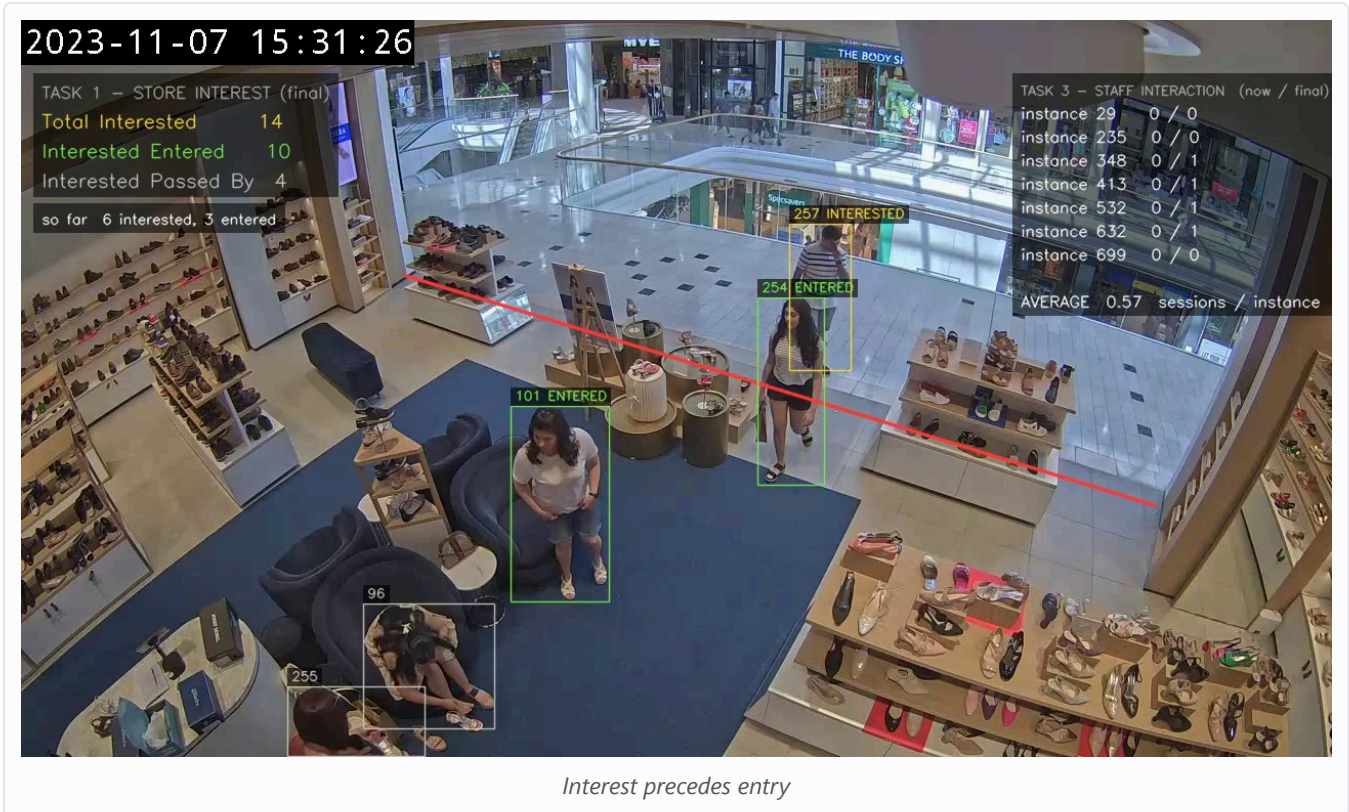
The threshold is 0.35, chosen by F1 rather than by Youden's J. Youden's J is what this project used until the zones were re-traced, and watching it move is what disqualified it: across three revisions of the shopfront polygon it jumped 0.346 -> 0.245 -> 0.528. With eight positives, the point maximising TPR - FPR sits on a step of the ROC that a single track can shift, and at 0.528 it collapsed `passed_by` to 1 -- destroying the interested-but-passed-by category the brief exists to measure.

enter	total	entered	passed by	conversion	precision	recall	F1
0.20	38	10	28	26 %	0.42	0.62	0.50
0.25	26	10	16	38 %	0.56	0.62	0.59
0.30	16	10	6	62 %	0.67	0.50	0.57
0.35	14	10	4	71 %	0.80	0.50	0.62
0.40	12	10	2	83 %	1.00	0.38	0.55

5.3 Entering: crossing is not enough



Track 565 is amber INTERESTED on the walkway side of the entrance line while 254 and 516 are green ENTERED inside it. Over her whole track she comes within 0.0003 m of the frontage and never crosses — proximity is not entry.



Track 257: the interest episode opens at 248.25 s and the store entry follows at 249.33 s. The two decisions are independent, and here they agree in the order one would hope for.

The brief says "cross the entrance boundary **and continue into** the store". So entry requires having been outside, then crossing, then still being inside 2 s later — **and reaching at least 0.5 m inside**.

The reason is that on the right of frame the walkway and the shopfront converge in the image, so a pedestrian walking straight past has a ground point that drifts a few pixels across the boundary polygon — which a pure containment test scores as entering the shop.

min depth	total interested	entered	passed by
0.0 m	14	11	3
0.3 m	14	10	4
0.5 m	14	10	4
0.7 m	14	10	4
0.9 m	14	10	4
1.2 m	12	8	4

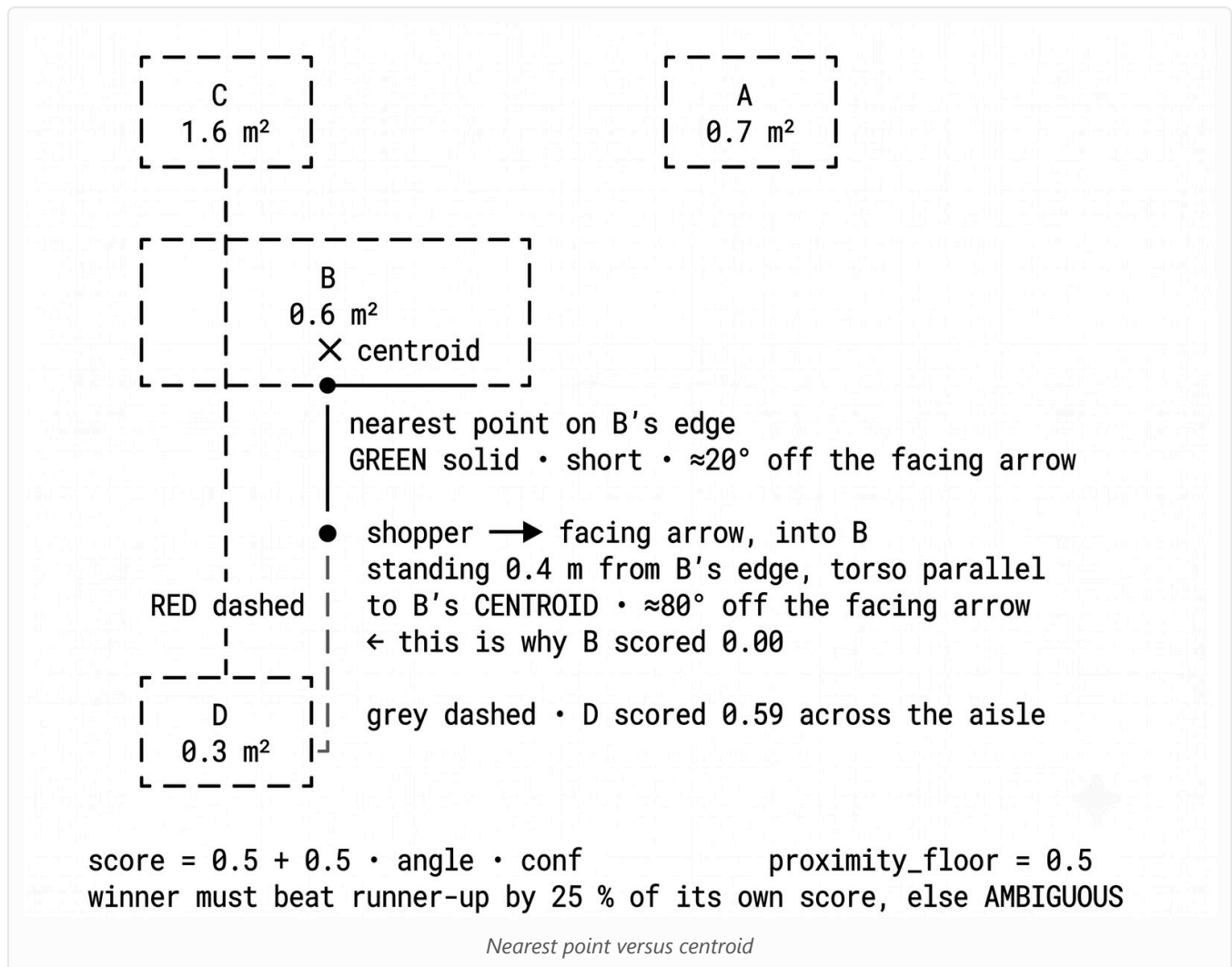
This parameter used to matter a great deal and now barely does, which is the more interesting result.

Under an earlier, badly traced store polygon the same sweep moved **entered** from 15 down to 2, and the depth requirement was carrying the whole burden of not counting pedestrians who brushed the boundary. Once the boundary was re-traced against the floor — section 6 describes the same class of error on the shelves — ground points stopped drifting across it and the patch stopped being load-bearing. It is kept because it is cheap and the failure it guards against is real on any scene where the boundary is less clean.

5.4 The partition holds by construction

Entering a shop *is* evidence of interest in it, and stronger evidence than any cue, so entry overrides the score. That makes `entered  $\subseteq$  interested` true by definition rather than by coincidence, and `passed_by` is the remainder. Two tests assert it.

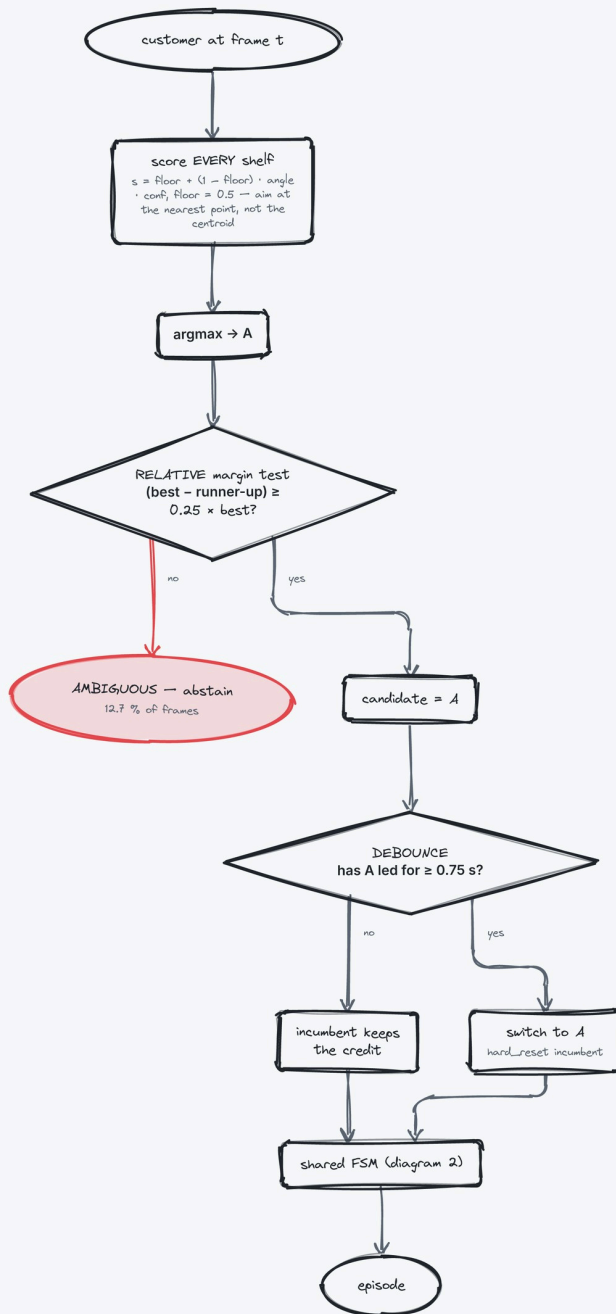
6. Task 2 - Per-shelf interest



Aim at the nearest point of the footprint, not its centroid. Containment survives projection; distance does not.

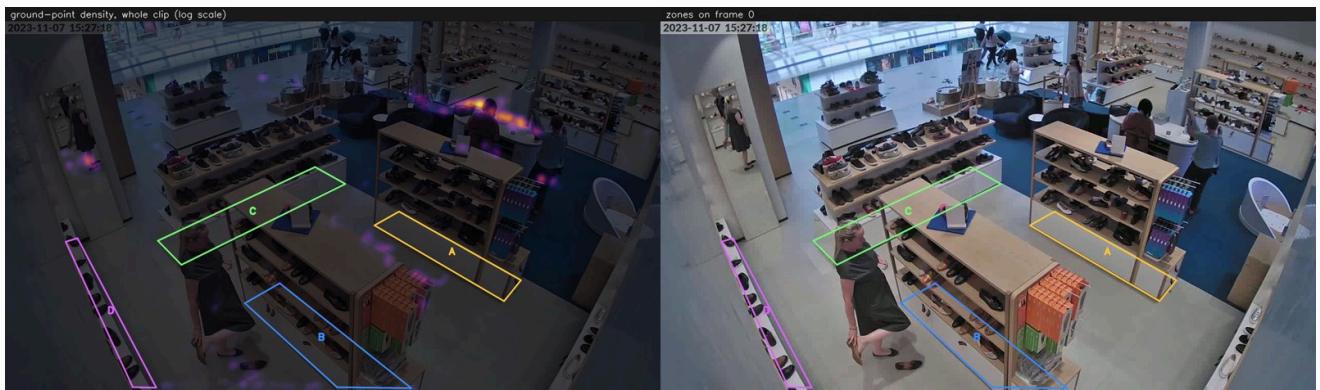
Per-frame shelf attribution

score → margin test → debounce → FSM



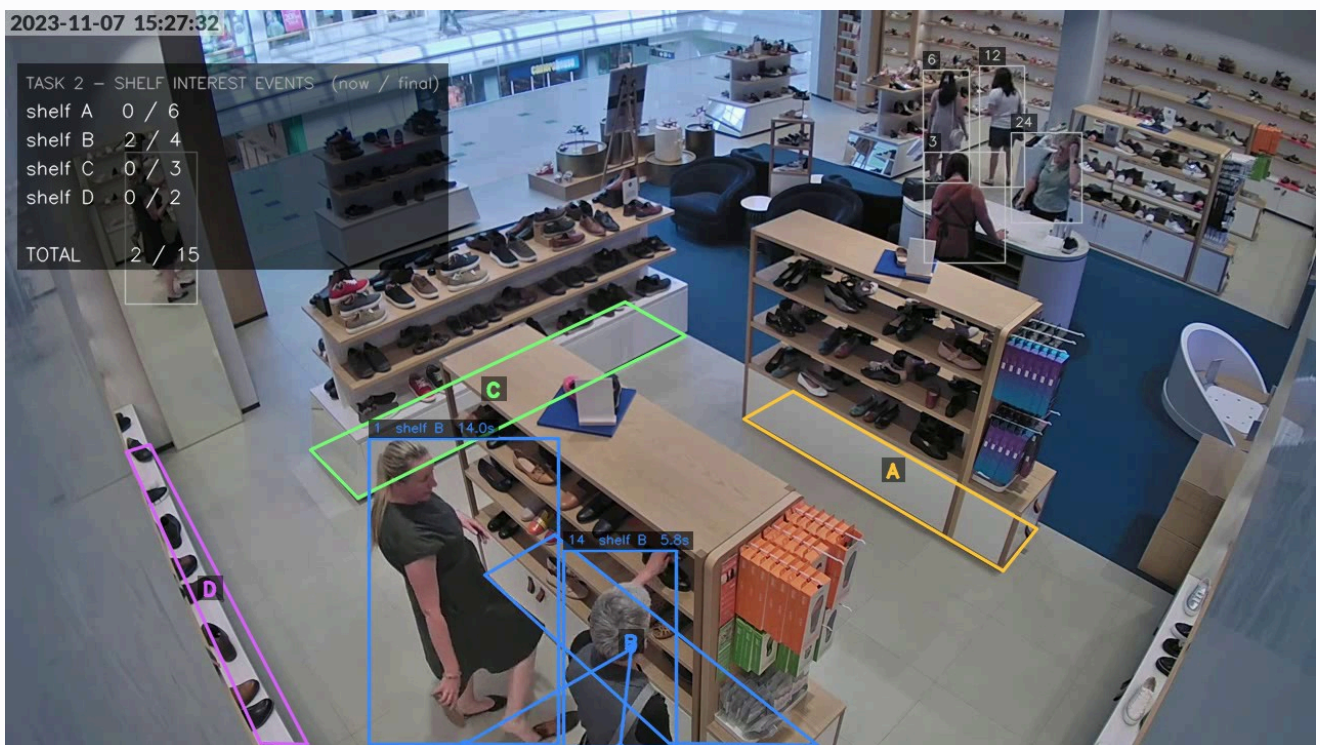
Per-frame shelf attribution

Score every shelf, take the argmax, abstain unless it wins by a relative margin, then require it to hold the lead for 0.75 s before the credited shelf actually changes. Three separate bugs lived on this path.



Shelf footprints over the interior ground-point density

The four shelf footprints are floor-contact quads, not the outline of each unit. A point a metre up a shelf projects to a floor position that does not exist.



Two customers, one shelf, both credited

Tracks 1 and 14 are both at shelf B and both credited to it, with live episode timers. Getting this right took three separate fixes — nearest-point aiming, a proximity floor, and a switch debounce.

2023-11-07 15:32:03

TASK 2 - SHELF INTEREST EVENTS (now / final)

shelf A	5 / 6
shelf B	3 / 4
shelf C ²⁰³	3 / 3
shelf D	2 / 2
TOTAL	13 / 15

181 202

200 shelf D 9.6s

A

B

C

TOTAL 13 / 15

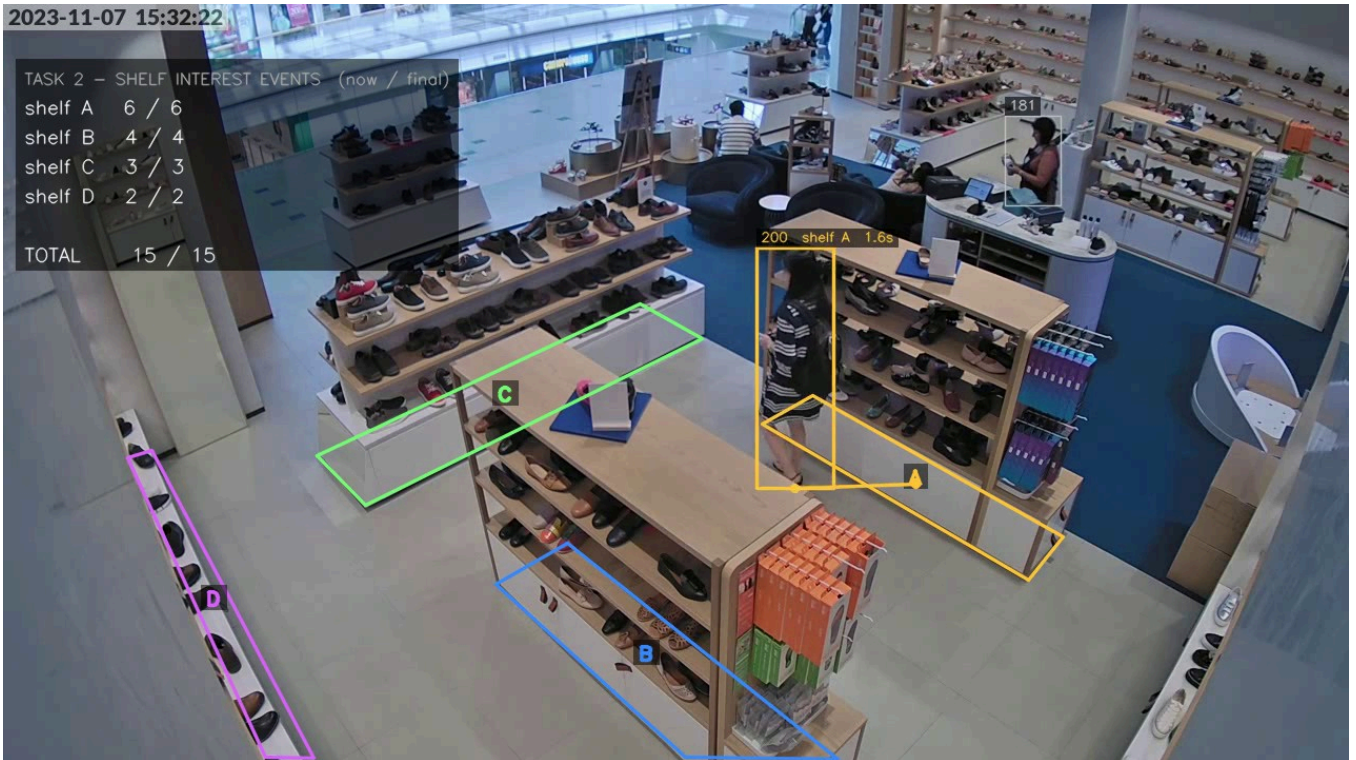
200 shelf D 9.6s

TASK 2 - SHELF INTEREST EVENTS (now / final)

shelf A	5 / 6
shelf B	4 / 4
shelf C	3 / 3
shelf D	2 / 2
TOTAL	14 / 15

TOTAL 14 / 15

200 shelf B 4.0s



Track 200 engages D (275.4–294.9 s), then B (295.0–302.8 s), then A (302.9–306.1 s). Three consecutive episodes, three separate events, no double-counting — which is the failure the brief names explicitly.

Per customer and per shelf, a score built from how close they are to the shelf and how much they are oriented toward it. Three separate defects were found here by tracing individual customers frame by frame against the video, and all three changed the answer.

Aim at the nearest point, not the centroid. The distance term used the nearest edge of the shelf's floor footprint while the angle term used its centroid. Standing 0.4 m from a 2.8 m gondola, the bearing to its middle is nearly perpendicular to where you are actually looking. A shopper handling stock at shelf B scored **B = 0.000** and **D = 0.59** -- D being a metre away across the aisle but roughly along her chest direction.

Proximity has to carry evidence on its own. The obvious form -- angle x confidence x distance -- lets orientation gate the whole score, so a customer scores zero on a shelf no matter how long they stand at it. That is wrong for the dominant behaviour in this clip: when you browse a fixture you stand *alongside* it and reach across, so your torso ends up parallel to it. The same shopper stood 0.4 m from shelf B for 28 seconds with her hands in its stock and was credited to a shelf 1.6 m away. Standing at a fixture for a sustained period is itself "evidence that they direct their attention toward it", so proximity now supplies half the score and orientation adds the rest.

A hold time before a switch counts. The margin below decides whether a single *frame* is decisive; it says nothing about whether a *change* is real. Without a hold, one frame where another shelf edged ahead switched the credited shelf, and because a switch hard-resets the incumbent's episode, **one visit was counted twice**. Customer 67 looked at shelf A continuously for eight seconds and was credited with two A events. That is precisely what the brief warns against.

Abstention instead of tie-breaking. At most one shelf is credited per frame. The winner must beat the runner-up by 25% of *its own score*; when it does not, nothing is credited and the frame is recorded as **ambiguous** -- a direct, quantified answer to the brief's "customer between two shelves" requirement. The

margin is relative, not absolute: shelf scores span two orders of magnitude, and a fixed gap called 56% of frames ambiguous even where one shelf outscored the next fourfold.

Floor-contact footprints, not shelf outlines. The floor model maps the ground plane, so a point a metre up a shelf projects to a floor location that does not exist. `tests/test_geometry.py` asserts the four footprints are disjoint.

Results: A 6 . B 4 . C 3 . D 2 = **15 events** from 11 distinct customers, with 12.7% of scored frames recorded as ambiguous.

Verification. There are no hand labels for Task 2 -- this is the least independently validated task in the submission, and that is stated rather than glossed over. What was done instead is direct visual spot-checking: five customers with long episodes were rendered with the shelf footprints drawn and checked by eye. All five are now correct; before the fixes, the one checked was wrong.

Sensitivity - reported honestly. The `min_duration` sweep shows no plateau, so the count is a judgement rather than a discovered constant; the whole curve is in `docs/sensitivity.md`. The debounce is now belt-and-braces rather than load-bearing: once proximity carries part of the score the flicker largely disappears at source, and the double-counting is zero at every hold time swept. It is kept because it costs nothing, and the flicker would return on a scene with shelves closer together.

7. Task 3 — Staff–customer interaction, and the staff classifier

7.1 Identifying staff — four approaches, and why the simplest won

The brief supplies the marker: staff "can be visually distinguished from customers by the aprons they wear". Ten staff instances were hand-verified from contact sheets (`labels/staff_candidates_entrance.csv`) — all ten are the **same apron-wearing employee**, appearing ten separate times.

Torso crops were cut from keypoints (shoulders-to-hips quad, asymmetric padding because the camera looks down, background blurred outside the torso polygon): 1,041 crops from 127 tracks.

method	track-level AP	recall at precision 1.0
HSV terracotta fraction (shipped)	0.933	0.70
CLIP linear probe, background-suppressed crops	0.900	0.60
CLIP linear probe, raw torso crops	0.883	0.70
colour + probe rank ensemble	0.967	0.70
zero-shot CLIP text prompts	0.506	0.20
CLIP appearance matching to confirmed staff	—	no separation

Chance is 0.079. **The learned model did not beat the interpretable baseline**, and the ensemble's higher AP does not change the operating point, so it would add a generalisation risk for no decision benefit. The colour rule ships.

7.2 Why the probe was rejected — the counterfactual

With one apron-wearing person in the clip, no train/test split can distinguish "learned the apron" from "learned *her*". Only intervening on the image can:

intervention	median score	change
staff, unmodified	0.797	—
staff, apron recoloured teal	0.822	+0.025
staff, torso blanked out entirely	0.668	−0.129
customer, unmodified	0.179	—
customer, torso painted terracotta	0.603	+0.424

A model reading the *garment* would collapse on the first two. Recolouring the apron away **did not lower the score at all**, while painting a customer's torso terracotta raised theirs by 0.42. So the model had two routes to "staff" — the colour and the employee's identity — and for the actual staff it was mostly using identity. That will not transfer to a different employee. This is exactly the failure an accuracy number at $n_{\text{positive_persons}} = 1$ cannot reveal.

Appearance matching failed for a related reason worth recording: crops taken from inside one shop are nearly collinear in CLIP space. Non-staff track cosine to the confirmed-staff centroid had median **0.937** and max **0.974**, while the staff instances the colour rule missed scored **0.855–0.951** — the negatives outrank the positives, so no threshold exists.

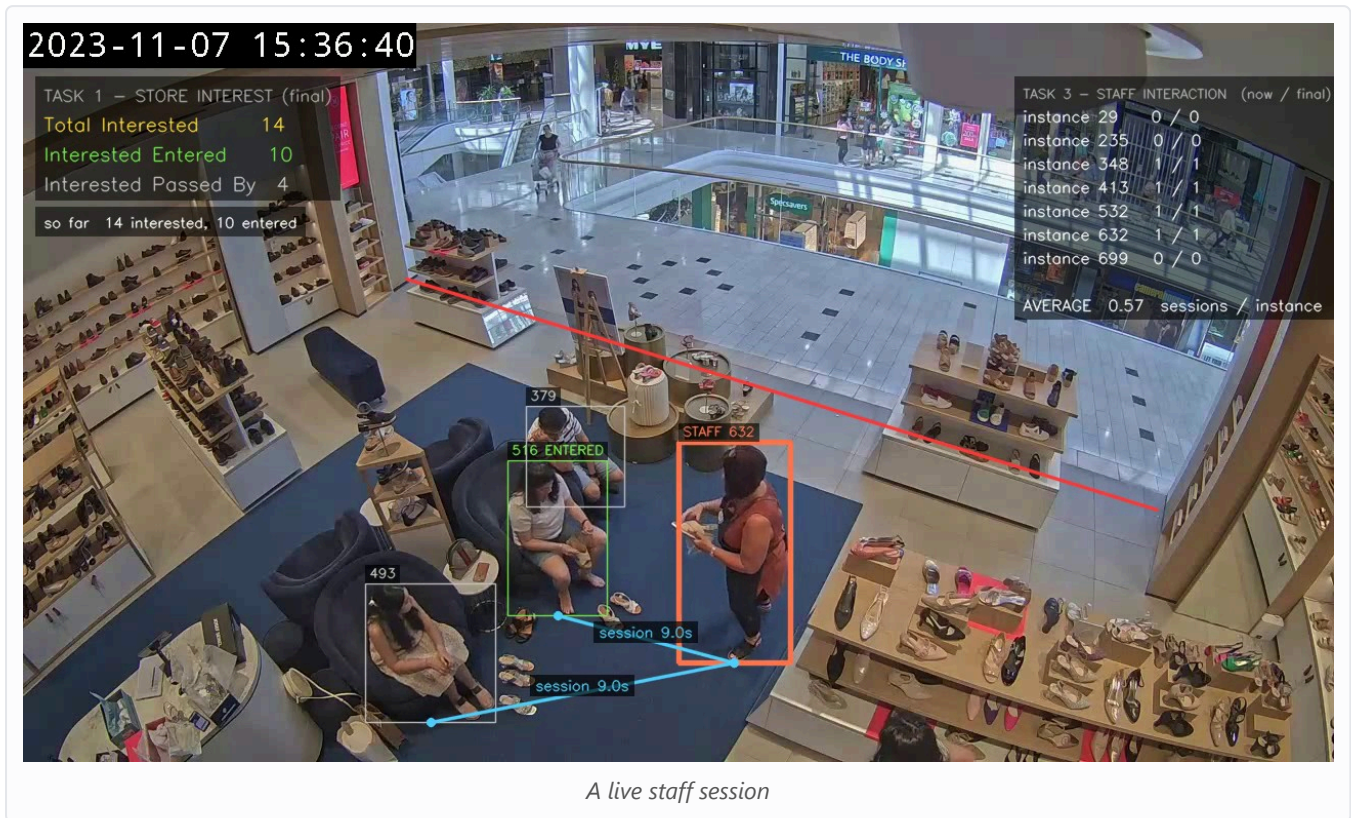
7.3 The shipped rule

Median apron fraction over a person's qualifying torso crops, requiring at least 8 of them, thresholded at 0.105.

- **Median**, measured against max / mean / p75 / p90 — the median won on every metric (AP 0.933 vs 0.811–0.925). The employee is often seen from behind where no apron is visible, and the median tolerates that; max rewards one lucky frame on a customer carrying something orange.
- **≥8 qualifying frames**. Without it, two-crop tracks of customers standing *beside* the employee inherit her colour and outrank real staff — tracks 52, 542 and 705 are exactly this, and are documented in the label file.
- **Threshold at precision = 1.0, maximum recall**. The costs are asymmetric: a false positive removes a real customer from Tasks 1 and 2 *and* adds a phantom instance to the Task 3 denominator, dragging the average down twice over.

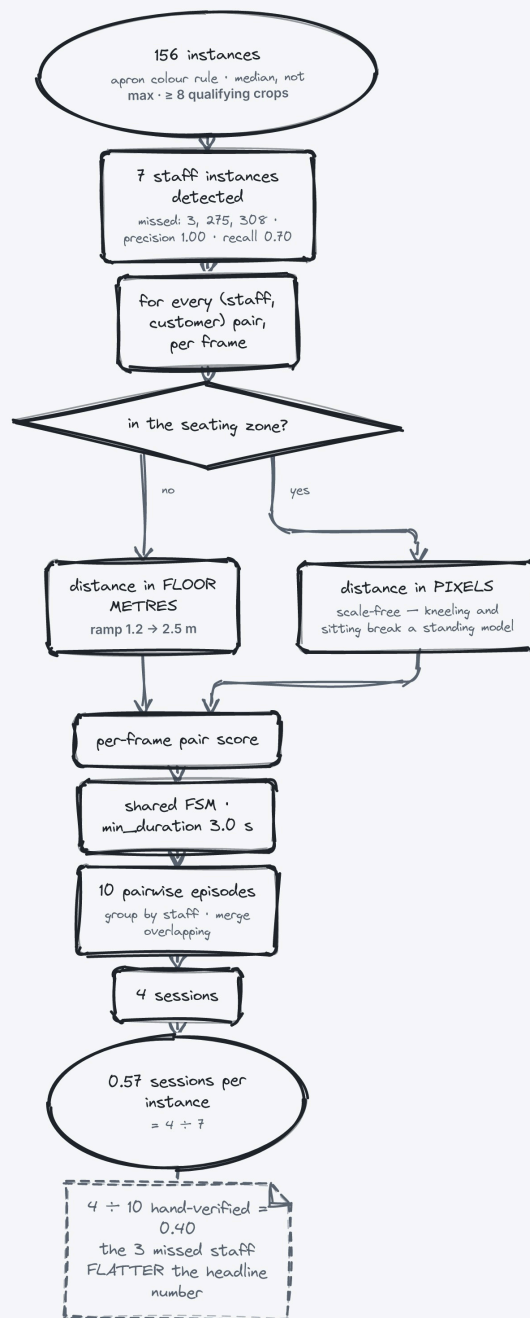
Result: **precision 1.00, recall 0.70** (7 of 10). Missed: tracks 3 and 275 (seen mostly from behind, median 0.092 and 0.060 against a 0.105 threshold) and track 308 (only 3 qualifying crops, so never scored at all).

7.4 Sessions



Staff 632 in orange with two live session links and their running durations, Task 1 counts top left, per-instance Task 3 totals and the 0.57 average right. The burnt-in clock is never drawn over: it is the ground truth the frame rate was measured against.

staff-customer interaction pipeline



From 156 instances to 4 sessions

The denominator is detected staff, not hand-verified staff. Three missed instances move the average from 0.40 to 0.57 — in the flattering direction, which is why it is stated here rather than left to be found.

An interaction is proximity that persists and is not someone walking past.

- floor distance, full score within 1.0 m decaying to nothing at 2.0 m (Hall's personal/social boundary ≈ 1.2 m plus the calibration's own error);
- a speed gate — a customer above 0.8 m/s is transiting, which is measured, not assumed: median walking speed here is 0.70 m/s and the p90 of per-track medians is 1.20 m/s;

- a **seating exception**: in the fitting area a staff member kneels and a customer sits, and both break a floor model that assumes standing. Inside the seating zone proximity switches to a scale-free *pixel* test (within 1.2–2.5× the larger box height), which needs no floor model at all.

Deliberately **not** required: that the two face each other. Staff kneel to fit shoes, and the label-free check in §4.4 puts the repaired estimator at 28° median on the entrance camera — good enough to weight a cue, not good enough to gate an interaction on. A requirement the evidence cannot support removes real interactions in exchange for false confidence.

Sessions, not pairs. A staff member serving a couple or a family is having one interaction, so overlapping pair-episodes for the same staff instance merge (5 s gap). 10 pairwise episodes → **4 sessions**.

The denominator is instances, per the brief: "a staff member should be treated as the same staff instance for as long as they remain within the camera view. If they leave the view and later re-enter, they may be treated as a new instance." Seven detected instances, zeros included → **0.57**.

The per-person diagnostic (all instances are one employee, so 4 sessions for one person) is reported in the events JSON and labelled as a diagnostic. It answers a different question from the one the brief asks.

8. Which parameters actually bind

`docs/sensitivity.md` re-runs the analytics across a grid for every threshold that could plausibly matter. Four results are worth pulling out.

Task 1 is bound by the entry depth, not by the interest threshold. `entered_min_depth_m` moves `entered` from 15 to 2 across the swept range. The interest threshold moves `passed_by` a great deal while leaving `entered` untouched -- which is the partition working as designed: entry is decided by geometry, and the interest score cannot interfere with it.

Task 2 is bound by `proximity_floor`, the parameter added last. It moves the count from 11 events (orientation-gated, 8 customers) to 21 (proximity only, 15 customers). 0.5 is the stated midpoint, and section 6 explains why neither extreme is right: pure orientation misses people browsing the shelf they stand at, pure proximity credits anyone who walks past one.

Task 3's minimum duration sits on a plateau two points wide. 3.0 and 4.0 s both give 4 sessions and 0.57; one step either side moves the answer — 2.0 s gives 5 sessions and 5.0 s gives 3. That is a narrower plateau than one would want, and on the evidence here it is the least well-supported threshold in the submission. It is called out rather than smoothed over because a reader checking `docs/sensitivity.md` will find it in thirty seconds.

`task3.group_merge_gap_s` does nothing at all. Completely flat across every value from 0 to 20 seconds. In this clip no two pair-episodes of the same staff instance ever fall within twenty seconds of each other without already overlapping, so the parameter never fires. It is reported *because* it is inert -- a carefully documented parameter that turns out to have no effect directs a reader's scrutiny at the wrong place. The distance gate and the minimum duration are what determine this answer.

9. Named failure cases

Full table with links to the review sheets in `docs/findings.md`.

Every disagreement with the hand labels, with a diagnosis.

Task 1 false positive (1). Track 127 — peak interest 0.58 with one sustained episode, on a label the reviewer marked *unsure*: a person who slows near the frontage while walking past, where the boundary between that and interest is genuinely fuzzy. It is the only false positive, and it disappears on the confident-only subset, which is why precision there is 1.00.

Task 1 false negatives (4). Tracks 65, 76 and 81 were originally labelled as *entering* the shop. Frame-by-frame review showed all three stay outside the shopfront and leave at the right edge of frame, so the labels were corrected to *passed by*, with the reason recorded in `labels/task1_labels.csv`. All three still carry an *interested* label the pipeline does not reproduce — peak scores 0.41, 0.32 and 0.71, none sustaining an episode. Track 115 is the clearest genuine miss: a 15 s zigzag along the frontage, repeatedly turning toward the window, peaking at 0.41.

Staff false negatives (3). Tracks 3 and 275 are the employee seen mostly from behind, where the apron is not visible; track 308 has only 3 qualifying crops and is never scored. Effect: the Task 3 denominator is 7 rather than 10, which *raises* the reported average ($4/7 = 0.57$ rather than $4/10 = 0.40$).

A staff-like population the brief's definition excludes. Several people in white tops kneel to fit shoes onto seated customers, which is staff behaviour. The brief defines staff by the apron, so they are counted as customers. Stated as a scope decision rather than an oversight.

Interior cadence below the human band (1.08 vs 1.6–2.4 steps/s). 54 % of interior ground points come from the box bottom rather than the ankles, because feet are occluded by fixtures, so step peaks are missed and intervals overestimated. The entrance clip, where 79 % of ground points are ankle-derived, lands inside the band.

10. Assumptions and limitations

Assumptions (stated, not measured here):

assumption	value	what it affects
adult standing height	1.70 m	absolute metric scale only
horizontal field of view	100°	absolute metric scale only
shoulder breadth / height	0.42 m / 82 % of stature	orientation estimate
camera optical axis is level and unrolled	—	the floor model
lens distortion is negligible	—	~1 % at frame edges; not corrected

Limitations:

1. **Orientation was wrong, and the first validation could not see it.** Both are fixed and documented in section 4.4, but the episode is the honest headline of this submission: a hand-labelled probe scored a

coin-flip estimator at 50 degrees because the labels and the estimator shared an author. Every validation here is now either label-free or explicitly flagged as not.

2. **The Task 1 interest labels are the candidate's own and unverified.** Twelve of 39 are marked unsure. Section 5.2 explains why the cue weights are no longer fitted to them.
3. **Task 2 has no hand labels at all.** It is the least independently validated task; five customers were spot-checked visually against the footage after the fixes and all five are correct, which is evidence but not a measurement.
4. **n_positive_persons = 1** for the apron classifier. The held-out score is not a confidence interval, it is a lower bound on uncertainty. Validation is by mechanism (section 7.2), not by sample size.
5. **Re-identification across separate appearances is scoped out**, as the brief permits. Tracker *id switches* while a person stays in view are repaired geometrically -- 23 repairs on entrance, 17 on interior, using only predicted position, apparent size and elapsed time, no appearance model. That is a different thing from re-ID and is argued as such in `src/rva/perception/relink.py`.
6. **The implied entrance camera height of 7.35 m is high for a mall.** Task 1 uses relative distances, so the absolute error largely cancels; it is reported rather than hidden.
7. **Interior orientation is markedly worse than entrance** -- 55 degrees median against 28, with 14% near-reversed against 5%. People inside a shop bend, crouch and reach, which breaks the upright-shoulders model the estimator assumes. This is why Task 2 leans on proximity.
8. **No plateau in the Task 2 `min_duration` sweep.** The chosen value is a judgement, and the whole sweep is published so a reader can pick differently.
9. **The rendered videos are not byte-reproducible.** x264 is not bit-stable across builds. The CSVs and event JSONs are.

11. Every threshold, tagged

`[measured]` derived from this footage · `[assumed]` a stated prior · `[fitted]` optimised against hand labels

parameter	value	tag
<code>timebase.frame_count / real_duration_s</code>	13421 / 600.0 → 22.368 fps	measured
<code>perception.conf</code>	0.25	measured (far-field people score 0.30–0.45)
<code>perception.imgsz</code>	960	measured (beat 640 and 1280 on far-field recall)
<code>perception.ankle_conf_min</code>	0.5	assumed
<code>relink.max_gap_s / max_dist_m</code>	1.0 s / 1.0 m	assumed (not swept)
<code>geometry.stature_m / hfov_deg</code>	1.70 m / 100°	assumed
<code>geometry.ruler.max_per_track</code>	40	measured (r 0.832 → 0.881)
<code>task1.min_observation_s</code>	2.5 s	assumed
<code>task1.reliable_* / near_*</code>	1.5–6.0 m / 0.3–1.5 m	measured (walkway is 2 m deep)
<code>task1.stop_max_cadence</code>	0.8 steps/s	measured (1.72 walking vs 0.39 stopped)
<code>task1.weights</code>	see section 5.2	fitted, (AUC - 0.5) per cue
<code>task1.fsm.enter</code>	0.35	fitted (F1; Youden's J rejected as unstable)
<code>task1.entered_min_depth_m</code>	0.5 m	fitted (swept, \$5.3)
<code>staff.min_box_h_px</code>	80 px	measured
<code>staff.min_frames_for_decision</code>	8	measured
<code>staff.decision_threshold</code>	0.105	fitted (precision = 1.0)
<code>task2.decide_margin</code>	0.25 relative	measured (an absolute margin left 56 % of frames ambiguous; relative gives 12.7 %)
<code>task2.fsm.enter</code>	0.30	derived
<code>task2.proximity_floor</code>	0.50	stated -- the binding one for Task 2
<code>task2.switch_hold_s</code>	0.75 s	measured (removes double counting)
<code>task2.dist_full_m / dist_zero_m</code>	0.0 / 2.5 m	measured
<code>task3.dist_full_m / dist_zero_m</code>	1.0 / 2.0 m	assumed (Hall)
<code>task3.customer_max_speed</code>	0.8 m/s	measured
<code>task3.group_merge_gap_s</code>	5.0 s	assumed

Unmeasured thresholds are `null` in the config and crash the pipeline rather than quietly producing a believable number. A misspelled config key fails at load, naming the field.

12. Output structure

outputs/	
entrance_annotated.mp4	Task 1 + Task 3, one video (as the brief requires)
interior_annotated.mp4	Task 2
task1_store_interest.csv	Total Interested / Entered / Passed By
task2_shelf_interest.csv	per-shelf event counts + total
task3_staff_interactions.csv	per staff instance (zeros included) + AVERAGE
events/	
entrance_events.json	every Task 1 and Task 3 decision, per track
interior_events.json	every Task 2 episode
calibration_*.json	the fitted floor model and its residual
staff_model_entrance.json	the classifier comparison and ablation
labels/	hand labels and the review sheets they came from
docs/figures/	timebase, calibration, zone previews
cache/tracks_*.parquet	Pass 1 output, committed for reproducibility

13. Where the evidence lives

claim	evidence
the frame rate is 22.368, not 30	<code>docs/figures/timebase_entrance.png</code> — the two clock readings
the floor model and its error	<code>docs/figures/calibration_*.png</code> — the fit, and a metre grid reprojected onto a frame
the zones are in the right place	<code>docs/figures/zones_*.png</code> — polygons over the ground-point density of the whole clip
pose orientation beats travel direction	<code>scripts/probe_orientation.py --scene entrance --selfcheck</code> — label-free. The hand labels in <code>labels/orientation_labels_entrance.csv</code> and <code>--score</code> are kept only as the record of a probe that could not tell 117° from 28° (§4.4).
the Task 1 hand labels	<code>labels/task1_labels.csv</code> + <code>labels/task1_sheets/*.jpg</code> (39 blind review sheets)
the staff hand-verification	<code>labels/staff_candidates_entrance.csv</code> + <code>labels/staff_sheet_entrance*.png</code>
the classifier comparison and ablation	<code>outputs/events/staff_model_entrance.json</code>
every threshold's sensitivity	<code>docs/sensitivity.md</code>
every disagreement with the labels	<code>docs/findings.md</code>